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Real-time sign language recognition using parallel multi-scale CNN to enhance inclusive education for deaf and hard of hearing students

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Abstract
Inclusive education, a fundamental right for all individuals, presents unique challenges in higher education, particularly when it comes to students with disabilities. Our research focuses on rectifying the gaps in catering to the needs of deaf university students by introducing real-time sign language recognition through a parallel multi-scale CNN. We conducted model testing with five different optimizers, including Adam, Nadam, Adadelta, RMSprop, and Stochastic Gradient Descent (SGD). Notably, the best test accuracy of 99.70% was achieved using SGD on the Sign MNIST dataset, which comprises 7200 images categorized into 24 alphabet classes (excluding 'J' and 'Z'). The parallel multi-scale CNN outperformed the conventional CNN model, which achieved an accuracy of 96%. Our approach outperformed other state-of-the-art methods, demonstrating superior performance based on accuracy, precision, recall, and F1-score. The model was tested in real-time scenarios with Moroccan deaf students pursuing degrees in electronics, computer science, and robotics at Mohammed V University in Rabat. This testing underscores the model's adaptability and its potential to improve inclusivity in higher education.

Keywords Inclusive education · Deaf students · Sign language recognition · Parallel multi-scale CNN · Deep learning

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1 Introduction

In today's educational landscape, the call for inclusive education has gained substantial momentum, largely driven by the United Nations Convention on the Rights of Persons with Disabilities (UNCRPD) Article 24 and its General Comment Number Four [36]. This influential declaration urges nations to swiftly establish fully inclusive education systems, grounded in the belief that segregating children with disabilities results in an education of subpar quality.

Within the realm of inclusive education, a diverse spectrum of individuals exists, each with its unique learning requirements and abilities. This encompasses students with a wide range of challenges, including students facing language barriers, visual impairments, cognitive difficulties, and physical disabilities, as well as students with hearing impairments who rely on sign language and various other conditions. In this environment, the goal is to create a setting where every learner, regardless of their abilities or disabilities, can thrive and access the same educational opportunities.

Inclusion is more than just the physical presence of students in general education classrooms, it prioritizes active engagement in a meaningful and challenging instructional program. Managing a classroom in inclusive education can present several significant challenges for teachers. One of the major challenges lies in the diversity of students' needs, which can vary significantly based on their abilities and specific requirements. Teachers must develop flexible and tailored lesson plans to address these diversified needs, which can be a complex and demanding task. Additionally, class size can be an issue as there may be a substantial number of students with special needs requiring individualized attention. This can lead to time and resource constraints as teachers strive to balance the needs of the entire class [24].

Communication is another major challenge. In an inclusive classroom, there can be a wide variety of communication modes, including sign language, communication aids, or augmented and alternative communication needs. Teachers must be prepared to use these different communication modes to ensure that each student has the opportunity to express themselves and understand the information presented in class. While some teachers excel in managing such diverse classrooms, the majority may face difficulties, often due to inadequate training and support [22]. Numerous traditional studies have employed information and communication technology (ICT) within the framework of inclusive education. Otta and Pozzi [28] emphasize the pivotal role of ICT in fostering e-inclusion and underscore the need for accessible tools and teacher competencies to achieve genuine inclusive education. Huang et al. [13] study on ICT education in rural Taiwan highlights the critical importance of adaptable teaching materials and localized methods in enabling successful information application learning in underserved communities. Drigas and Ioannidou [10] emphasize the transformative role of ICTs in special education, highlighting their capacity for more comprehensive assessment and effective interventions, ultimately enhancing the learning experiences of students with special educational needs.

Recent studies have explored inclusive education by harnessing the advancements in artificial intelligence, machine learning, and deep learning to better support students with disabilities. Srivastava [32] underscores the role of AI-driven Smart Learning Assistance tools in promoting inclusive education, ensuring equitable access, and effective communication for students with disabilities within the classroom. Won et al. [15] show the value of deep learning algorithms in predicting school absenteeism among autistic students, potentially enhancing their access to essential educational and therapeutic support while minimizing the

risk of dropout. Ketcham [18] highlights the promising potential of the intelligent glove in aiding individuals with visual impairments by detecting obstacles and providing real-time location data, setting the stage for future enhancements in improving its functionality and user experience. Lasri et al. [23] show the potential of deep learning and transfer learning in facial expression recognition to evaluate engagement levels of deaf and hard-of-hearing students, offering a promising approach for enhancing classroom experiences and assessing student engagement more effectively.

In a related context, recent studies in sign language recognition have made significant advancements, particularly in enhancing communication for deaf and hard-of-hearing students. Utilizing machine learning and deep learning techniques, researchers have developed systems that can translate sign language gestures into text or spoken language, bridging the communication gap between sign language users and those unfamiliar with sign language. These developments hold great promise for promoting inclusivity and breaking down communication barriers, ultimately enriching the educational experience for individuals with hearing and speech impairments. Notably, Bin Munir [2] underscores the importance of improved communication between deaf-mute and non-mute individuals through an intelligent sign language interpretation system. Additionally, Kothadiya [20] presents a deep learning-based algorithm employing long short-term memory (LSTM) and gated recurrent units (GRU) networks, offering a promising solution to enhance communication for those with impaired speech or hearing by accurately detecting and recognizing sign language gestures. Furthermore, Pigou's [29] research leverages technology, including Microsoft Kinect and convolutional neural network (CNN), to automate sign language recognition, offering a promising solution to improve communication between the deaf community and the hearing majority. According to these studies, it is evident that there are numerous obstacles in the field of inclusive education, especially concerning the integration of sign language to ensure effective communication.

In response to these challenges, we have developed a model based on parallel multi-scale CNN to comprehensively address these obstacles. Notably, our approach stands out as a unique contribution, as there is currently no existing work that employs our specific model in this context, and we are pioneers in applying this innovative model in the context of inclusive higher education. The motivation behind this research stems from the urgent need to enhance inclusive education, particularly for deaf and hard-of-hearing students. Despite technological advancements, significant barriers to communication and learning access persist in inclusive classrooms. Teachers often face challenges in meaningfully engaging all students, resulting in disparities in educational access and outcomes. This research aims to bridge this gap by leveraging innovative technologies, specifically a parallel multi-scale convolutional neural network model, to improve real-time sign language recognition. We aspire to empower educators and create an inclusive learning environment where all students can thrive.

Furthermore, we conducted rigorous testing using five different optimization techniques, including Adam, Nadam, stochastic gradient descent (SGD), Adadelta, and Rmsprop. We applied the model to deaf and hard-of-hearing students pursuing inclusive bachelor's degrees in electronics, computer science, and robotics at the Faculty of Sciences, Mohammed V University in Rabat, Morocco, to ensure its effectiveness and adaptability. Our model aims to revolutionize the education system by introducing a cutting-edge solution for seamless communication between teachers and students through sign language. We recognize the importance of inclusive education, and our innovative approach leverages technology to bridge the communication gap. This model will empower teachers to learn and effectively use sign language, breaking down barriers and ensuring that students who rely on sign language have equal access to quality education. By incorporating this solution into classrooms, we

strive to create an inclusive learning environment where all students, regardless of their communication abilities, can fully engage.

The primary aims of this investigation can be summarized as follows:

- We propose a novel real-time deep learning sign language model based on parallel multi-scale CNN.
- We conducted evaluations using various model optimizers on the Sign MNIST dataset to assess the model's efficacy in sign language recognition.
- We validate our model to monitor the real-time effectiveness of sign language communication within an inclusive bachelor's degree program, which includes deaf and hard-of-hearing students, for the first time.

The rest of this article is described as follows: Section 2 presents the related works in the field of sign language recognition. Section 3 contains our proposed methodology based on parallel multi-scale CNN. Section 4 discusses the experimental results. Finally, Section 5 presents the conclusion and outlines the future extensions of our work.

2 Related work

Sign language recognition systems have been developed using diverse approaches, including artificial neural networks (ANNs) and convolutional neural networks (CNNs), each aiming to address the challenges posed by variable backgrounds, lighting conditions, and computational complexity. Various feature extraction techniques have been explored to enhance recognition accuracy across different sign languages and alphabets. Yasir et al. [37] adopted support vector machines for Bengali sign language recognition, leveraging shift-invariant feature transform (SIFT) and k-means clustering. Similarly, Jin et al. [7] employed SVMs for American Sign Language (ASL) recognition, using Canny edge detection and seeded region growing for hand gesture segmentation. Hartanto et al. [12] and Huong et al. [13] experimented with HSV color space segmentation, achieving varying degrees of success in Indonesian and Vietnamese sign language recognition. Notably, Jimoh et al. [16] designed an Android-based system using OpenCV template matching for English vocabulary recognition, emphasizing potential improvements through different feature extraction techniques. Kulkarni and Lokhande [21] utilized artificial neural networks and Otsu's segmentation, coupled with Canny edge detection for feature extraction. While achieving decent accuracy, they struggled with computational demands and sensitivity to background variations.

In efforts to enhance accuracy and address the challenges, advanced techniques like Histogram of Oriented Gradient (HOG) and CNNs have been applied. Tabassum et al. [33] used Canny edge and HOG for ASL alphabet recognition, while Masood et al. [25] used very deep convolutional networks (VGG16) to achieve impressive results with the Massey dataset [4]. Tolentino et al. [35] focused on beginners' ASL learning using a CNN and skin-color technique, attaining notable accuracy rates. Dudhal et al. [11] introduced a CNN-based Indian sign language recognition system, employing hybridized adaptive thresholding and SIFT techniques. Wadhawan and Kumar [26] extended this approach with grayscale and color images, achieving high training accuracy. Das et al. [9] and Jain et al. [14] utilized CNNs to recognize American Sign Language and highlighted the significance of filter size, demonstrating the potential for improvements.

Mittal et al. [26] designed a sign language recognition system using LSTM and Leap Motion. They applied a four-gated LSTM cell with a 2D CNN for sentence recognition, achieving 72.3% accuracy. Individual word recognition attained an accuracy of 89.50% with

this system. Aparna and Geetha [3] propose a pioneering sign language recognition method, blending CNNs and LSTM networks. They use pre-trained CNNs for feature extraction, followed by LSTM layers, achieving a notable 94% accuracy on the Indian Sign Language (ISL) dataset, demonstrating its significance in deep learning-based sign language recognition.

Despite these advances, significant challenges persist, particularly regarding background interference, illumination variations, and computational complexity. Existing methods often struggle to accurately recognize signs in diverse real-world environments, where background clutter can obscure hand movements and lighting changes can lead to misinterpretations. Additionally, many models face high computational demands, hindering their deployment in real-time applications. Our study addresses these issues by utilizing a robust architecture designed to effectively manage background variability and illumination fluctuations. We also incorporate optimized feature extraction techniques that enhance accuracy while reducing computational requirements. By employing a standardized dataset, we aim to fill gaps in the literature and enable clearer comparisons across different sign languages, ultimately improving the reliability and effectiveness of recognition systems.

3 Proposed methodology

In this section, we will address key elements, including the proposed system architecture for real-time sign language recognition using a parallel multi-scale CNN, model optimization strategies, and the evaluation of the model's performance through various metrics.

3.1 Proposed system architecture for real-time sign language recognition using parallel multi-scale CNN

Figure 1 presents the flowchart for the proposed model, outlining the overall process of the multi-scale convolutional neural network. Following this, Fig. 2 details the architecture of the multi-scale convolutional neural network, which is designed with parallel branches. Within the convolutional block, three branches utilize 3x3, 5x5, and 7x7 convolution kernels, respectively. Each branch applies a rectified linear unit (ReLU) activation function to

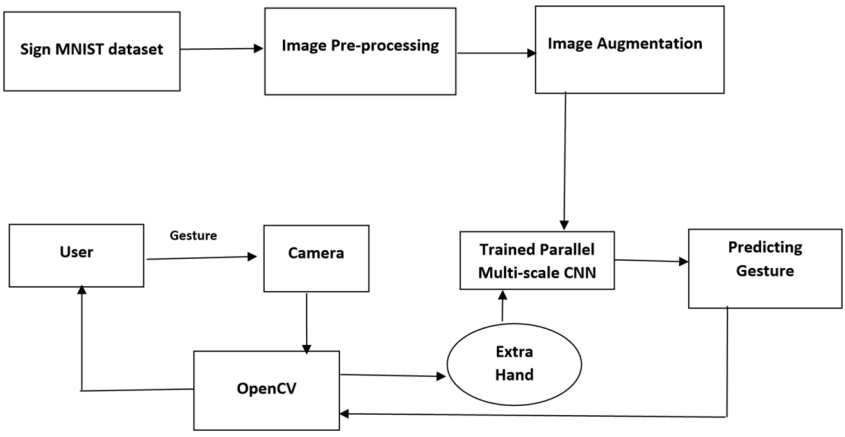


Fig. 1 Flowchart for the proposed model

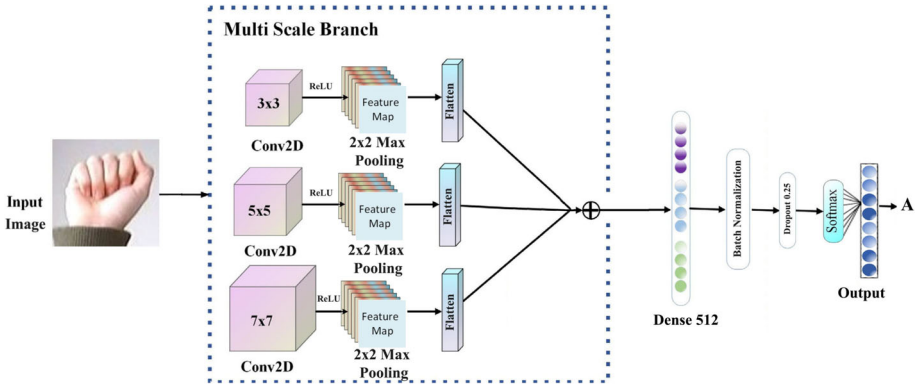


Fig. 2 Generic architecture of the parallel multi-scale CNN for sign language recognition

introduce nonlinearity, as shown in (1). This multi-branch architecture enables the model to effectively capture diverse spatial patterns across multiple scales.

$$R(x) = \begin{cases} max(0, x) & , x >= 0 \\ 0 & , x < 0 \end{cases} \tag{1}$$

Afterward, a max-pooling layer with a 2x2 kernel size and a stride of 2 is employed for down-sampling. The flattened layers play an important role in converting multidimensional feature representations extracted from each frequency band into one-dimensional feature vectors.

The results obtained from the various convolution kernel branches are combined, capturing the spatial-temporal connectivity characteristics of the four frequency bands. This fusion result is then fed into fully connected layers. The first layer consists of 1000 neurons, while the second layer has 512 neurons. The output layer comprises 24 neurons, aligned with the number of classes being predicted, facilitating the computation of predicted probabilities for each letter class. A Softmax function is applied to the output layer, as described in (2).

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \tag{2}$$

where C represents the number of different classes, the subscript i denotes the ith class, and z_i is the predicted score for the ith class.

After the fully connected layers, rectified linear activation functions (ReLU) are applied, followed by batch normalization layers and dropout layers with a dropout probability of p = 0.1 to prevent overfitting. Figure 3 provides a summary of our parallel multi-scale CNN model architecture. The algorithm below details the steps involved in this model.

3.2 Model optimization

To manipulate and optimize the categorical cross-entropy loss function of our model we used five optimization algorithms namely Stochastic Gradient Descent (SGD) (Robbins & Monro [30]), ADAM (Kingma & Ba [19]), Nadam (Nesterov [27]), Adadelta (Zeiler [38]), and RMSprop (Tieleman & Hinton [34]). The categorical cross-entropy loss function is

Layer (type)	Output Shape	Param #	Connected to
input_3 (InputLayer)	[(None, 28, 28, 1)]	0	[]
conv2d_6 (Conv2D)	(None, 26, 26, 32)	320	['input_3[0][0]']
conv2d_7 (Conv2D)	(None, 24, 24, 32)	832	['input_3[0][0]']
conv2d_8 (Conv2D)	(None, 22, 22, 32)	1600	['input_3[0][0]']
max_pooling2d_6 (MaxPooling2D)	(None, 13, 13, 32)	0	['conv2d_6[0][0]']
max_pooling2d_7 (MaxPooling2D)	(None, 12, 12, 32)	0	['conv2d_7[0][0]']
max_pooling2d_8 (MaxPooling2D)	(None, 11, 11, 32)	0	['conv2d_8[0][0]']
flatten_7 (Flatten)	(None, 5408)	0	['max_pooling2d_6[0][0]']
flatten_8 (Flatten)	(None, 4608)	0	['max_pooling2d_7[0][0]']
flatten_9 (Flatten)	(None, 3872)	0	['max_pooling2d_8[0][0]']
concatenate_2 (Concatenate)	(None, 13888)	0	['flatten_7[0][0]', 'flatten_8[0][0]', 'flatten_9[0][0]']
dense_4 (Dense)	(None, 512)	7111168	['concatenate_2[0][0]']
batch_normalization_2 (BatchNormalization)	(None, 512)	2048	['dense_4[0][0]']
dropout_2 (Dropout)	(None, 512)	0	['batch_normalization_2[0][0]']
dense_5 (Dense)	(None, 24)	12312	['dropout_2[0][0]']

Total params: 7128280 (27.19 MB)
Trainable params: 7127256 (27.19 MB)
Non-trainable params: 1024 (4.00 KB)

Fig. 3 Model summary of the parallel multi-scale CNN

represented as follows:

$$CE = -\sum_{i}^{C} t_i \log\left(f(s)_i\right) \tag{3}$$

Where C corresponds to the total number of distinct classes, while the subscript i references the ith specific class. Within this framework, t_i denotes the actual truth label, S_i signifies the Softmax probability associated with the ith class, and z_i represents the predicted score for that particular class.

- Stochastic Gradient Descent (SGD) [30]: replaces the actual gradient vector with a Stochastic (random) estimate of the gradient vector. This stochastic estimate refers to the gradient of the loss function for a single data point or instance and is represented with the following formula:

$$\theta = \theta - \eta \nabla J(\theta, x^{(i)}, y^{(i)}) \tag{4}$$

Algorithm 1 Parallel Multi-Scale CNN Algorithm.

Input: Grayscale image of size (28, 28, 1).
Step 1: Apply three convolutional layers in parallel:
Convolution 1: 3x3 kernel.
Convolution 2: 5x5 kernel.
Convolution 3: 7x7 kernel.

$$(I * K_{NxN})_{i,j} = \sum I_{i+m,j+n} \cdot K_{m,n}, \quad N = \{3, 5, 7\}$$

Step 2: Apply ReLU activation to the output of each convolution:

$$\text{ReLU}(x) = \max(0, x)$$

Step 3: Apply 2x2 MaxPooling to the feature maps:

$$P_{i,j} = \max_{m,n \in \{0,1\}} \left\{ F_{2i+m,2j+n} \right\}, \quad F = \{F_{3x3}, F_{5x5}, F_{7x7}\}$$

Step 4: Flatten the output from each MaxPooling layer:

$$F_{flattened} = \text{Flatten}(P_{3x3}), \text{Flatten}(P_{5x5}), \text{Flatten}(P_{7x7})$$

Step 5: Concatenate the flattened outputs:

$$\text{Concatenation} = [F_{flattened,3x3}, F_{flattened,5x5}, F_{flattened,7x7}]$$

Step 6: Apply a dense layer with 1000 units, ReLU activation, Batch Normalization, and Dropout (0.1).
Step 7: Apply another dense layer with 512 units, ReLU activation, Batch Normalization, and Dropout (0.1).
Step 8: Apply a dense layer with 24 units and Softmax activation:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Output: A probability distribution over 24 classes.

Where θ represents the model parameters that we aim to optimize, while η denotes the learning rate controlling the step size during parameter updates. The function $J(\theta, x^{(i)}, y^{(i)})$ serves as the cost function, measuring the model's error for a given training example $(x^{(i)}, y^{(i)})$. Additionally, $\nabla J(\theta, x^{(i)}, y^{(i)})$ corresponds to the gradient of the cost function concerning the parameters θ for that specific training example.

- Adaptive Moment Estimation (ADAM) [19]: is a commonly used method. The moment the update is converted into an exponential moving average. The equation of the ADAM optimization algorithm is represented as follows:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) \nabla J(\theta_t) \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2)(\nabla J(\theta_t))^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \\ \hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \\ \theta_{t+1} &= \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \end{aligned} \tag{5}$$

θ represents the model parameters we want to optimize. α is the learning rate, which controls the step size when updating the parameters. m and v are the first and second moments (moving averages) of the gradients. t is the current iteration step. β_1 and β_2 are the exponential decay rates for the first and second moments, typically close to 1. ϵ is a small constant to prevent division by zero.

- Nesterov Adam (Nadam) [27]: combines the techniques of Nesterov Accelerated Gradient (NAG) and Adaptive Moment Estimation (Adam) into a single optimization algorithm.
- Adadelta [38]: is a variant of SGD that introduces adaptability in learning rates for each dimension. It aims to mitigate two primary issues encountered in traditional gradient descent: the gradual reduction of learning rates during training and the necessity for manual selection of a global learning rate. Adadelta represents a more resilient advancement over Adagrad, as it dynamically adjusts learning rates by considering a moving window of gradient updates instead of accumulating all past gradients. This unique approach allows Adadelta to maintain effective learning, even in scenarios where numerous updates have already taken place.
- RMSprop [34]: maintains a decaying average of squared gradients and normalizes the current gradient by the square root of this average. The centered variant further calculates the average of gradients to estimate variance.

3.3 Model evaluation

In the domain of classification studies, the assessment of deep learning models frequently involves the utilization of performance metrics derived from the confusion matrix. This matrix serves as a visual representation that illustrates the correspondence between the predicted class labels generated by the model and the actual labels assigned to input images.

True Positive (TP) instances occur when the deep learning classifier accurately identifies an image as displaying the intended sign, correctly predicting a sign language gesture when the image is labeled as such. Conversely, False Positive (FP) cases arise when the classifier incorrectly identifies an image as depicting a different sign. On the other hand, False Negative (FN) situations happen when the deep learning model of the classifier wrongly predicts an image as representing a different sign. Lastly, True Negative (TN) scenarios occur when an image is correctly predicted as not representing a particular sign by the classifier's deep learning model.

We utilize a set of four performance metrics, denoted as (6), (7), (8), and (9), which encompass:

- Accuracy: these metrics gauge the proportion of correct predictions (TP and TN) relative to the total number of predictions made.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

- Precision: evaluates the model's ability to make accurate positive predictions and is calculated as TP divided by the sum of TP and FP.

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

- Recall: measures the model's capacity to identify all relevant instances and is computed as TP divided by the sum of TP and FN.

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

- F1 Score: is the harmonic mean of precision and recall, offering a balanced measure of a model's performance

$$F1 = 2\text{x}\frac{(Precision\text{x}Recall)}{(Precision + Recall)} \tag{9}$$

4 Experimental results and discussion

In the upcoming subsections, we will begin with the experimental setup, providing insights into the configuration and parameters employed. Following that, we will delve by explaining the dataset used in our study. Afterward, we will thoroughly examine and present the outcomes obtained through the application of the proposed model. Finally, we will discuss the results of real-time sign language recognition for deaf and hard-of-hearing students, emphasizing the practical implications of our research.

4.1 Experimental setup

We employed the Python-based Keras library [8] to facilitate the integration of models with different layer counts into the training workflow. The input images were resized to dimensions of 224 × 224 pixels before being fed into the models. We used the parallel multi-scale CNN model. The entire execution took place within the Google Colab environment, utilizing GPUs to ensure consistent conditions for all four models. Throughout the training phases, we monitored model weights to capture the highest validation accuracy.

In the process of training our deep learning model, we applied several data augmentation techniques designed to enhance the diversity and robustness of our training dataset. Firstly, we rescaled the pixel values of our images to a range of [0, 1] to ensure consistency in data representation. We then introduced randomness through various transformations. While we didn't apply image rotation, justifying this decision by the fact that it doesn't significantly affect the interpretation of alphabet characters in a meaningful way, we did employ other transformations. We allowed for random shifts in the height and width dimensions of the images, with a maximum shift of 20%. Furthermore, we applied zooming to our images, randomly zooming in by up to 20%, providing the model with different scales of the same objects. Additionally, horizontal flipping was enabled, which mirrors images horizontally, adding further diversity. Lastly, in the case of any new pixels created during these transformations, we used the 'nearest' pixel to fill them, ensuring the integrity of the image. Table 1 provides a summary of the optimal hyperparameters utilized during the training process. Following training, various performance metrics were assessed to evaluate the effectiveness of the models.

4.2 Sign language MINST

We used the Sign Language MNIST dataset sourced from Kaggle [17]. It consists of 34627 images of hand signs with 24 classes of letters (excluding J and Z, which involve motion)

Table 1 Optimal hyperparameters for the parallel multi-scale CNN model

Parameter	Value
Input image dimension	(224, 224, 3)
Activation function	Softmax
Loss function	Categorical Cross-Entropy
Learning rate	0.1
Momentum	0.9
Decay	1e-4
Batch size	200
Training epochs	100

and in which every training and test instance corresponds to a label ranging from 0 to 25, meticulously aligned with each alphabetic letter from A to Z. It's noteworthy that there are no instances for the letters J (9) or Z (25) due to the inherent nature of gesture motions associated with these letters. In total, the dataset comprises 34627 cases, comparatively, the size of the training set encompasses 27455 cases, while the test set consists of 7172 cases.

The Sign Language MNIST dataset follows a CSV format, with a header row containing a label, followed by pixel values from pixel1 to pixel 784. These pixel values meticulously encode a single 28x28 pixel image, capturing grayscale intensities that range between 0 and 255. This pixel-level representation adeptly captures the visual essence of each image, forming the foundation for subsequent analysis and recognition processes. Figure 4 provides an example of images from the Sign Language MNIST dataset.



Fig. 4 Example of images from the sign language MNIST dataset

Table 2 Performance metrics of CNN and parallel multi-scale CNN models

Model	Accuracy	Precision	Recall	F1-score
CNN	96%	99%	98%	99%
Parallel multi-scale CNN	99.70%	100%	99%	100%

4.3 Experimental results

In this study, we have compared the performance of our proposed model parallel multi-scale CNN with CNN. Using four evaluation metrics including accuracy, precision, recall, and f1-score, as shown in Table 2 and Fig. 5, parallel multi-scale CNN stands out as the top performer with an outstanding accuracy rate of 99.7%. While the CNN model reaches 96%.

The performance comparison of different optimizers, as displayed in Table 3, reveals insightful findings based on test accuracy. Notably, the stochastic gradient descent (SGD) optimizer outperforms the others with a remarkable test accuracy of 99.70%. The Adam optimizer also delivers a competitive performance with a test accuracy of 95.63%, Furthermore, the Nadam optimizer achieves a test accuracy of 82.50%, followed by the Adadelta optimizer at 76.50%, and the RMSprop optimizer with a test accuracy of 72.40%. These results provide valuable insights into the impact of different optimization algorithms on the model's performance, with SGD emerging as the most effective choice in this particular context. This performance difference can be attributed to the fact that SGD is particularly effective for image-based tasks, such as sign language recognition, where spatial patterns are important. In comparison, optimizers like Adam, Nadam, Adadelta, and RMSprop, which perform better in tasks with sparse gradients, such as text classification, struggle to capture the necessary details in image recognition tasks.

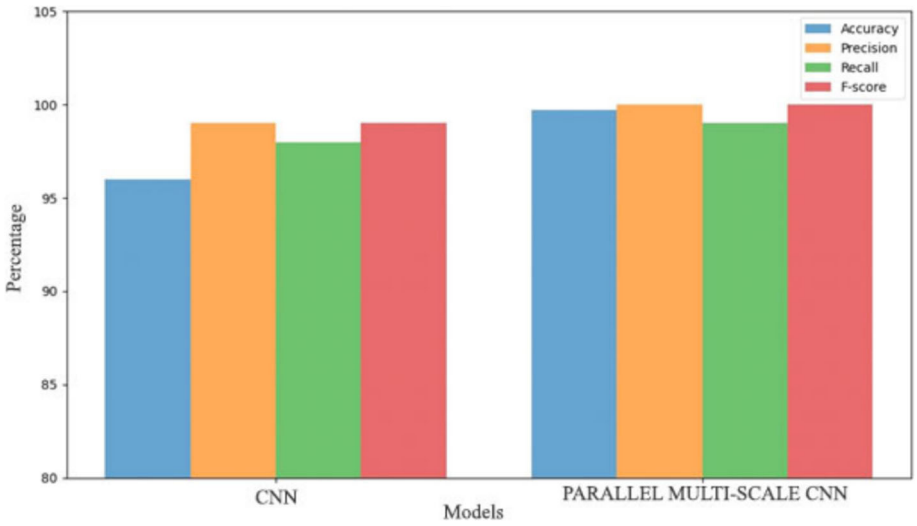


Fig. 5 Comparison of various performance metrics for convolutional neural network (CNN) and parallel multi-scale CNN models

Table 3 Performance comparison of optimizers based on test accuracy

Optimizer	Test accuracy
SGD	99.70%
Adam	95.63%
Nadam	82.50%
Adadelta	76.50%
RMSprop	72.40%

Figure 6 shows the confusion matrix of our model and Fig. 7 illustrates the learning curve of the parallel multi-scale CNN model with SGD optimizer over the course of 100 epochs, showcasing the evolution of training and test accuracy in (a), as well as the corresponding training and test losses in (b).

Table 4 provides a comprehensive overview of various methods for classification and recognition, highlighting their individual performances. The 2D-CNN and modified LSTM with a Leap Motion sensor, used in [26], achieved an accuracy of 89.50% on a custom dataset

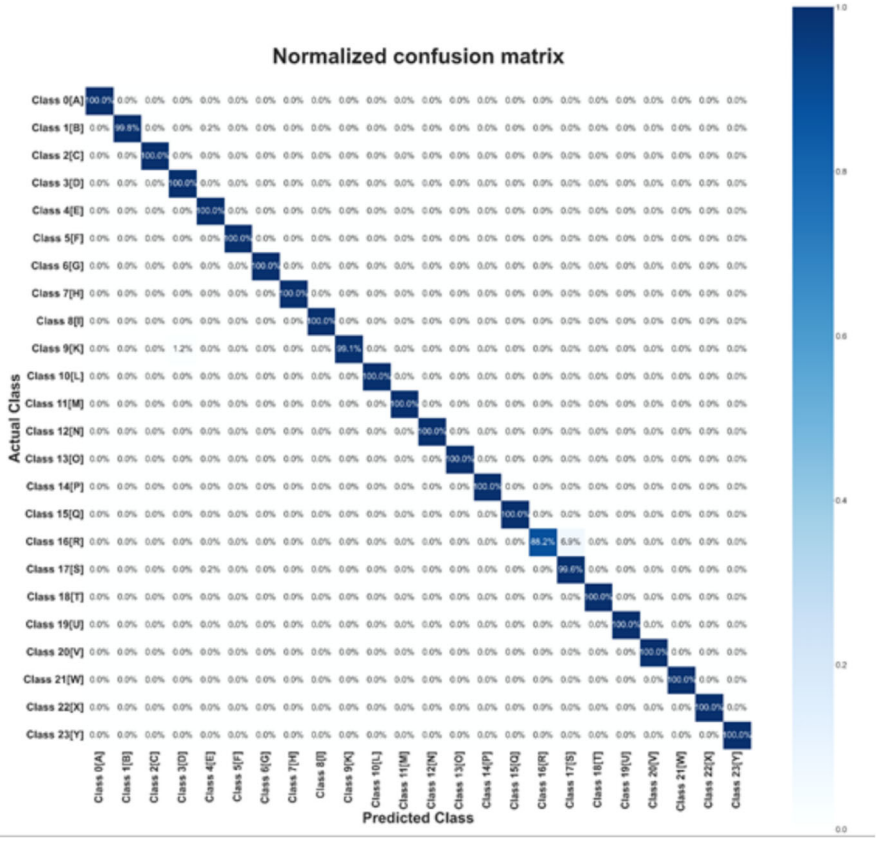


Fig. 6 Confusion matrix for the parallel multi-scale CNN with SGD optimizer

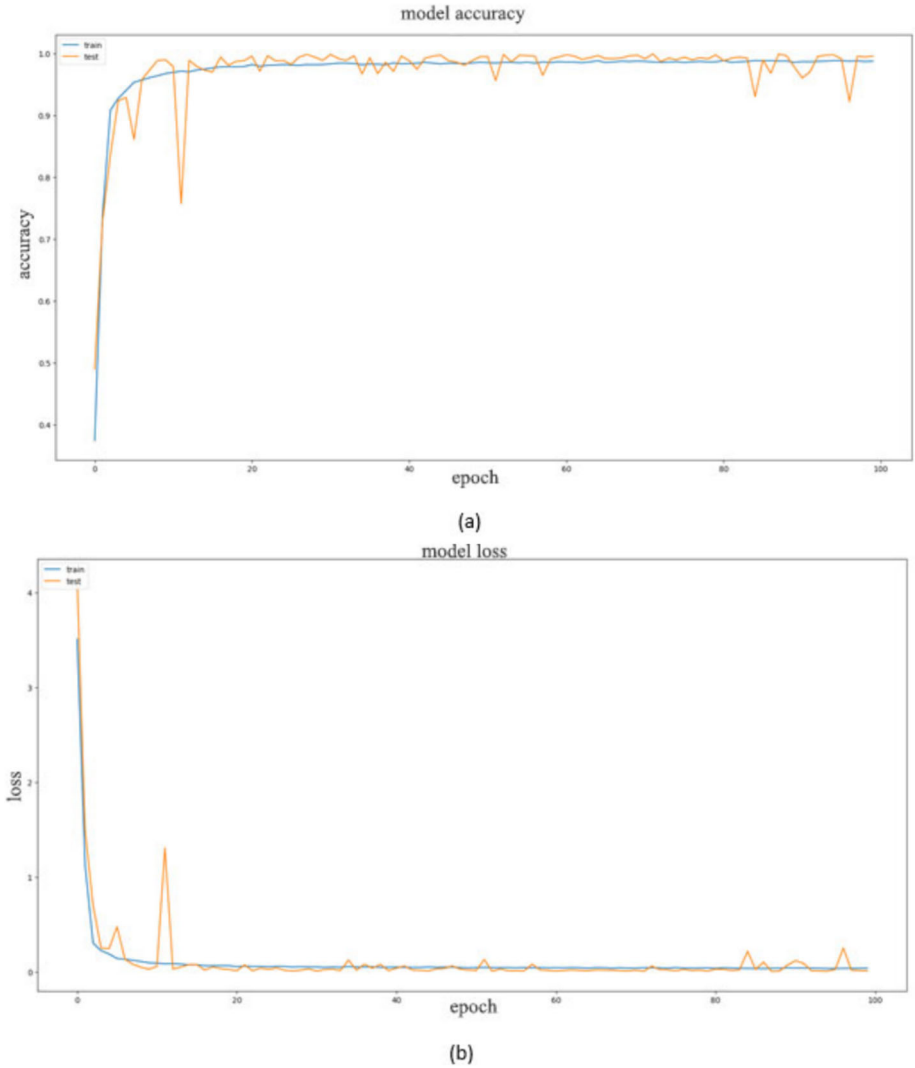


Fig. 7 Training and testing accuracy (a) and loss (b) of the parallel multi-scale CNN with SGD optimizer

of 3,150 ASL images. The CNN [1] applied to the NUS Hand Posture dataset demonstrated an accuracy of 94.7%, with a precision of 94.96%, recall of 94.85%, and an F1-score of 94.26%. In handwritten digit classification on the MNIST dataset, augmented CapsNET [31] reached an accuracy of 95.08%. A CNN-LSTM-based model [3] attained 94% accuracy using a custom dataset of 20 videos related to Indian sign language, while a KNN model [5] achieved an accuracy of 91% on a custom dataset of 780 ASL images. In comparison, a YOLOv5 model [6] applied to the Ecuadorian Sign Language dataset, consisting of 1602 images, achieved a precision of 85.5%, accuracy of 92%, and recall of 85.7%.

Table 4 Comparison of sign language recognition accuracy among different approaches

Methodology	Dataset	Language	Accuracy	Precision	Recall	F1-Score
2D-CNN and Modified LSTM with Leap motion sensor [26]	Custom Dataset (3150 images)	American	89.50%	–	–	–
YOLOv5 model [6]	Ecuadorian Sign Language dataset (1602 images)	Ecuadorian	92%	85.5%	85.7%	–
KNN [5]	Custom Dataset (780 images)	American	91%	–	–	–
CNN and 2 layer LSTM [3]	Custom Dataset (20 videos)	Indian	94.00%	–	–	–
CNN [1]	NUS hand Posture dataset	American	94.7%	94.96%	94.85%	94.26%
CapsNET augmented [31]	MNIST dataset	American	95.08%	–	–	–
Our proposed approach	MNIST dataset	American	99.70%	100%	99%	100%

In contrast, our proposed parallel multi-scale CNN model stands out with an exceptional accuracy rate of 99.70% on the Sign MNIST dataset, alongside a precision of 100%, recall of 99%, and an F1-score of 100%. This remarkable accuracy can be attributed to the use of three different convolution kernels (3x3, 5x5, and 7x7) in parallel within the convolutional block, enabling the model to capture diverse spatial patterns across multiple scales and providing a richer feature extraction process. The 3x3 kernel captures fine details, while the 5x5 and 7x7 kernels detect larger, more global patterns. Furthermore, the ReLU activation function applied after each convolution introduces non-linearity, allowing the model to learn complex and varied feature representations. This combination of multi-scale feature extraction and non-linear activation enables the model to generalize effectively, achieving high accuracy by adapting to both small and large variations in sign language gestures. Unlike some more complex architectures, our method leverages the power of multi-scale feature extraction while maintaining simplicity. Our approach, compared to single-kernel CNN models or more intricate designs, enhances performance without overcomplicating the model, demonstrating that simpler models can yield superior results in this specific context.

4.4 Results of sign language real-time recognition for deaf and hard-of-hearing students

We conducted a real-time sign language recognition experiment at the Faculty of Sciences in Rabat Mohammed V University, Morocco, to assess our model's performance for deaf and hard-of-hearing students pursuing degrees in licence of electronics, computer science, and robotics. This experiment occurred in the Ibn Al-Baytar amphitheater and involved four participants, one male and three females, with two of them using hearing aids. We obtained their consent after providing a detailed explanation of the study's objectives, the methodology, and the use of their hand gestures for research purposes. Figure 8 illustrates that the parallel multi-scale CNN model effectively detected and accurately identified the hand signs executed by the participants, facilitating clear communication between the participants and the system. The hand signs were precisely delineated by black outlines, highlighting the model's capability to isolate the regions of interest (the hand gestures) from the surrounding image, an important aspect for precise recognition. Our model provides invaluable support to educators by enhancing real-time sign language recognition in the classroom, which significantly improves the quality of interaction between teachers and deaf or hard-of-hearing students. By facilitating seamless communication, the system eliminates the traditional barriers that often hinder full participation in classroom discussions, lectures, or activities.

The integration of this technology not only ensures effective communication but also promotes active student engagement, empowering deaf students to take part in the learning process on equal footing with their peers. From a pedagogical perspective, this model helps educators adapt their teaching strategies to be more inclusive and responsive to the needs of students with hearing impairments. Teachers can now deliver lessons without relying solely on interpreters, allowing for more spontaneous and dynamic exchanges between them and their students. Additionally, the system can support the implementation of differentiated instruction, where educators can tailor their teaching methods to accommodate different learning styles and abilities, thereby fostering a more personalized and effective educational experience.

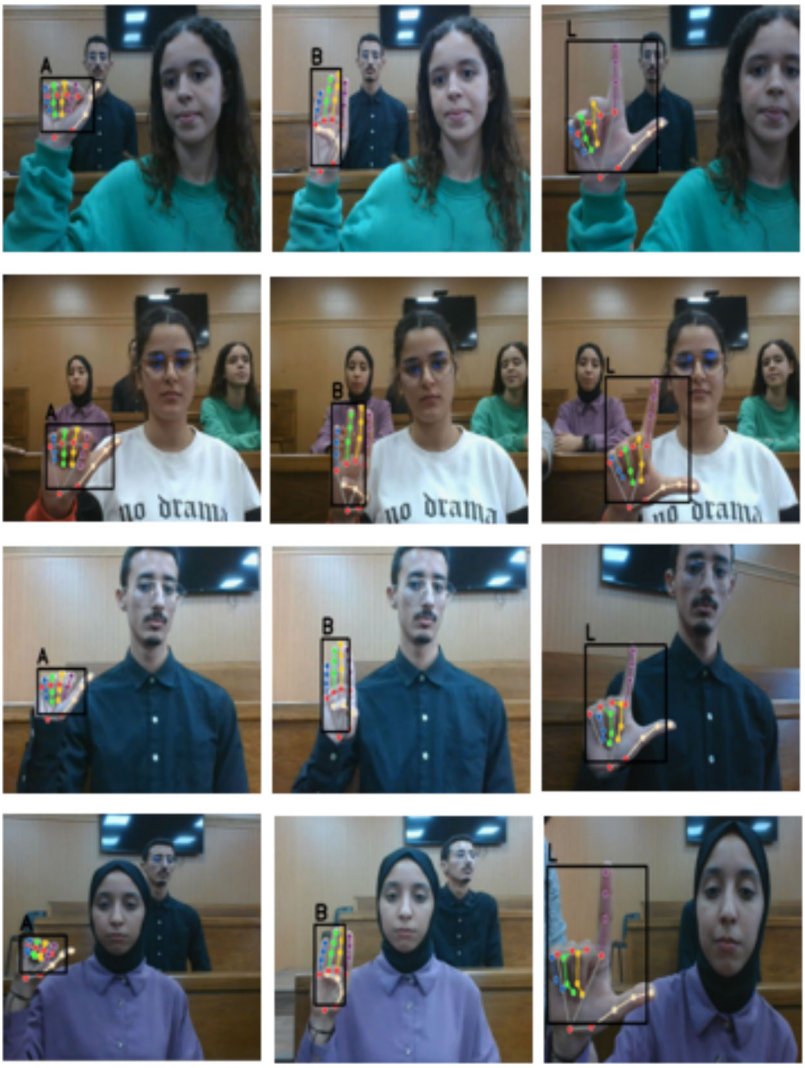


Fig. 8 Real-Time sign language recognition of deaf and hard-of-hearing student using parallel multi-scale CNN model

5 Conclusion

Our research underscores the paramount significance of inclusive education. In this context, we have developed a model based on a parallel multi-scale CNN to overcome the challenges hindering the effective integration of sign language. The multi-scale convolutional neural network in our study employs a parallel architecture with three branches that utilize 3x3, 5x5, and 7x7 convolution kernels respectively and ReLU activation functions. A max-pooling layer reduces the dimensions of feature maps, while a Flatten layer transforms multi-dimensional features into one-dimensional vectors. The outcomes from these branches are combined to capture spatial-temporal connectivity across four frequency bands. The fully connected

layers encompass a 1000-neuron layer, a 512-neuron layer, and a 24-neuron output layer with Softmax activation for class prediction, with each class representing an alphabet letter (excluding 'J' and 'Z'). Additional ReLU activation, batch normalization, and dropout layers with a 0.1 probability enhance learning and mitigate overfitting. We conducted rigorous testing with five different optimization techniques, including Adam, Nadam, SGD, Adadelta, and Rmsprop, to determine the optimal optimizer.

Our findings affirm the effectiveness of our parallel multi-scale CNN model, which achieved an outstanding accuracy rate of 99.70% on the MNIST dataset using SGD with a learning rate of 0.1, a momentum of 0.9, decay of 1e-4, a batch size of 200, and 100 training epochs. Our proposed model surpassed state-of-the-art methods. We applied our model to both typically hearing students and those who are deaf or hard of hearing, pursuing an inclusive bachelor's degree in electronics, computer science, and robotics at the Faculty of Sciences of Mohammed V University in Rabat, Morocco, ensuring its efficacy and adaptability. Our model stands as a unique contribution, with no comparable work in this field. We are pioneers in the application of this innovative model within the realm of inclusive higher education. By introducing this model into classrooms, our aim is to foster an inclusive learning environment that enables all students, regardless of their means of communication, to actively engage in the educational process.

While our model successfully addresses many challenges, it is important to note the absence of the letters 'J' and 'Z' from the dataset, as these require dynamic motion for accurate representation. We plan to address this limitation in future work by developing solutions that employ sensors to capture and enhance the dynamic movements associated with these letters. Furthermore, we are committed to expanding our research to Moroccan Sign Language, which will broaden the scope and impact of our work in creating inclusive educational environments tailored to diverse sign language communities.

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Data availability statement The Sign MNIST dataset from the following link is used in this work. https://www.kaggle.com/datasets/datamunge/sign-language-mnist

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Compliance with Ethical Standards Informed consent was obtained from all individual participants involved in the study.

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